Math Geometry Formula Cheat Sheet

Essential 2D & 3D Formulas for Middle & Early High School Math (Grades 7–10)

VARIABLE LEGEND

- r = Radius
- d = Diameter ($d = 2r$)
- h = Height
- b = Base length
- l, w = Length, Width
- s = Side length
- l = Slant height (3D)
- B = Area of 3D Base
- P = Perimeter of 3D Base

1. 2D SHAPES (PLANE GEOMETRY)

SHAPE	AREA (A)	PERIMETER (P) / CIRCUMFERENCE (C)	KEY NOTES
Square	$A = s^2$	$P = 4s$	All 4 sides equal (s)
Rectangle	$A = l \cdot w$	$P = 2l + 2w$	Opposite sides equal
Triangle	$A = \frac{1}{2} b h$	$P = a + b + c$	Height (h) is perpendicular to base (b)
Parallelogram	$A = b \cdot h$	$P = 2a + 2b$	Height must be perpendicular to base
Trapezoid	$A = \frac{1}{2} (a + b) h$	$P = a + b + c + d$	a and b are parallel bases
Circle	$A = \pi r^2$	$C = 2 \pi r$ (or $C = \pi d$)	Use $\pi \approx 3.14159$ or $^{22}/_{7}$

2. 3D SHAPES (SOLID GEOMETRY)

SHAPE	VOLUME (V)	TOTAL SURFACE AREA (SA)	ADDITIONAL FORMULAS / NOTES
Cube	$V = s^3$	$SA = 6s^2$	6 identical square faces
Rectangular Prism	$V = l \cdot w \cdot h$	$SA = 2lw + 2lh + 2wh$	Box shape with 6 rectangular faces
General Prism	$V = B \cdot h$	$SA = 2B + P \cdot h$	B = base area, P = base perimeter
Right Cylinder	$V = \pi r^2 h$	$SA = 2\pi r^2 + 2\pi r h$	Lateral Area (LSA) = $2\pi r h$
Right Cone	$V = \frac{1}{3} \pi r^2 h$	$SA = \pi r^2 + \pi r l$	Slant height: $l = \sqrt{(r^2 + h^2)}$
Square Pyramid	$V = \frac{1}{3} s^2 h$	$SA = s^2 + 2s l$	Base is square ($s \times s$), 4 triangular faces
Sphere	$V = \frac{4}{3} \pi r^3$	$SA = 4 \pi r^2$	Hemisphere Volume = $\frac{2}{3} \pi r^3$

QUICK MEMORY & PROBLEM SOLVING TIPS

- **Prism vs. Pyramid Volume:** A pyramid or cone is always ⅓ the volume of a prism or cylinder with the same base and height ($V_{pyramid} = \frac{1}{3} V_{prism}$).
- **Surface Area Concept:** Surface Area is simply the total sum of the areas of all 2D faces surrounding the 3D shape.
- **Perpendicular Height:** Always make sure height (h) is measured perpendicular (90°) to the base, not along a slanted edge (unless finding lateral area with slant height l).